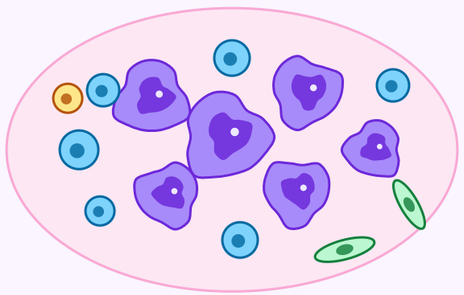
Do single-cell foundation models help identify malignant cells?

Zero-shot benchmark of scGPT, Geneformer and UCE against simple baselines and random controls on tumour scRNA-seq

1 Tumour scRNA-seq



Tumour cells + tumour microenvironment

76 patients · 7 datasets · 7 cancer types

Melanoma · HNSCC · LUAD · Breast · CRLM · Glioblastoma · Foll. lymphoma

Malignant cell

Non-malignant cell

Malignant vs all normal cells, and vs lineage-matched normal cells

2 Embed (zero-shot)

Foundation models

scGPT	Geneformer	UCE
512-d	768-d	1280-d

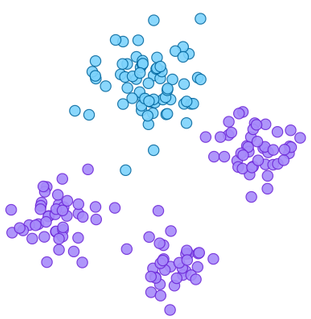
Baselines (*Harmony uses patient labels)

HVG + PCA	Harmony*
50-d	50-d

Negative controls

Random proj.	scGPT, random
512-d	512-d, untrained

no fine-tuning · identical cells for every model



cell embedding space (schematic)

3 Evaluate

T1 Unsupervised
Leiden clusters & kNN purity
no labels used

T2 Held-out patient probe
leave-one-patient-out; full or 10/50 cells per class per patient; label noise 0–20 %

T3 Cross-cancer transfer
train on 6, test on held-out type; 1 study per cancer type (= leave-one-study-out)

T4 Patient structure
normal and malignant cells scored separately

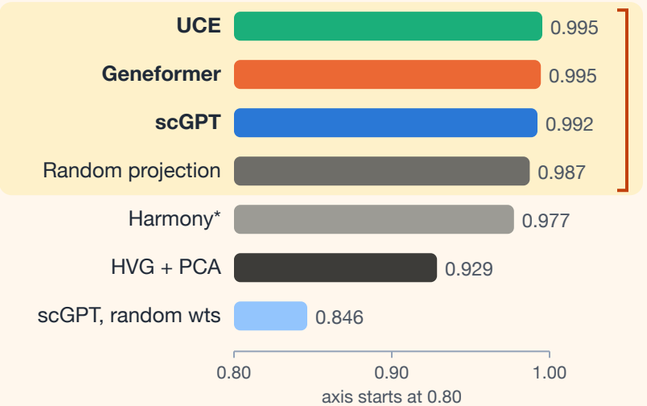
Patient = unit of replication

patient-bootstrap 95 % CI · paired Wilcoxon · Holm correction

4 Key findings

10-shot probe AUROC, mean of 76 held-out patients

10 training cells per class per patient · no label noise



Random projection recovers 88%–92% of the FM gain over HVG + PCA; FMs keep a small, significant edge (+0.005 to +0.008, Holm p < 0.001)

1. **Standard task is saturated**
   malignant vs all normal, full probe: AUROC ≈ 0.99 for FMs and baselines, incl. random projection
2. **Cross-cancer transfer splits the models**
   UCE highest label-free (mean 0.94, tied with Harmony*); Geneformer inverted on lymphoma (0.44, one study)
3. **Patient effects persist**
   normal-cell patient structure: FMs 0.36–0.43, HVG + PCA 0.40 (technical batch or real biology)
4. **Label noise erodes the edge**
   10 % flipped labels: random projection 0.815, FMs 0.808–0.827